Inequalities — problem 7.78

A. Engel — Problem-Solving Strategies · 1998

Find the positive solutions of the system of equations

$$x_1 + \frac{1}{x_2} = 4,\ x_2 + \frac{1}{x_3} = 1, \ldots, x_{99} + \frac{1}{x_{100}} = 4,\ x_{100} + \frac{1}{x_1} = 1.$$